

Robert Charles

Senior Data Engineer

508 Vann Rd, Highlands, TX 77562 · 2547089501 · worlddev8081@gmail.com

An accomplished Senior Data Engineer with 9 years of experience architecting and implementing data solutions within the Microsoft ecosystem. Specializes in data model design, handling large-scale data integrations, and optimizing cloud data warehouses using tools like Azure Data Factory, Azure Databricks, and Spark. Expertise in building secure, scalable, and automated data pipelines, including the use of OAuth 2.0 for API integrations. Demonstrated ability to mentor teams, enhance data quality management, and lead cross-functional collaborations to improve business outcomes. Proficient in delivering technical solutions and documentation for broad project scopes.

WORK EXPERIENCE

Senior Data Engineer

August 2024 – January 2026

Corelight, Remote

- Architected an enterprise-wide data strategy using Azure Synapse and Azure Data Factory, integrating data from 20+ sources, improving data ingestion speed from 1 hour to 15 minutes.
- Engineered a secure data lake architecture on Azure Data Lake, automating data quality checks with Databricks, reducing validation time by half.
- Developed a high-volume event streaming platform with Azure Event Hubs and Apache Kafka, processing 2M events per day with real-time insights.
- Coded a RESTful API integration framework using OAuth 2.0 for third-party data import, normalizing JSON formats for seamless SQL translation.
- Simplified data warehouse design using Dimensional Modeling techniques, enhancing report generation performance from 30 minutes to 3 seconds.
- Created SQL-based transform logic, dramatically reducing query runtime from 20 seconds to 10 seconds on average.
- Integrated Great Expectations for automated data validation, identifying 500+ discrepancies monthly across datasets.
- Mentored a team of 5 junior engineers, increasing their productivity and skill proficiency on cloud platforms by implementing a comprehensive onboarding program.
- Led cross-departmental workshops and training sessions, fostering data-driven decision-making across the organization.
- Documented technical architectures and workflows, establishing a centralized knowledge base that streamlined project on-boarding for new team members significantly.
- Coordinated with stakeholders to prioritize and deliver on data-intensive projects, consistently meeting tight deadlines.
- Standardized monitoring and logging across all data pipelines using Azure Monitor, decreasing incident response time from 45 minutes to 5 minutes.

Data Engineer

February 2021 – June 2024

Augury, Remote

- Constructed an intelligent data orchestration platform using Azure Data Factory and Apache Airflow, enhancing data flow efficiency, cutting execution time from 3 hours to 20 minutes.
- Streamlined data movement and transformation of 1TB of data daily with SQL Server Integration Services (SSIS), improving data availability.

- Established a scalable Azure Databricks environment to process real-time analytics across 10,000 IoT devices, optimizing throughput from 1000 to 30,000 events per second.
- Implemented a metadata-driven pipeline using Python and Databricks Delta, ensuring data governance and lineage tracking.
- Automated ingestion processes for JSON APIs using Azure Logic Apps, minimizing manual intervention during third-party data integrations.
- Formulated and executed a redundancy elimination strategy enhancing database integrity for 50 million records, achieving zero data duplication.
- Pioneered an automated error detection system, leading to a reduction in data synchronization issues across multiple environments.
- Delivered monthly business analytics dashboards using Power BI integrated with Azure Synapse, enabling 500 end-users to make informed strategic decisions.
- Redesigned the ETL pipeline architecture for enhanced modularity, significantly improving maintainability and deployment speed.
- Coordinated with business units to derive precise data requirements, ensuring seamless data integration aligned with business objectives.

Data Analyst

February 2018 – January 2021

Lob, Remote

- Refactored legacy data workflows into a cohesive AZURE environment, substantially reducing processing times from 2 hours to 20 minutes.
- Executed a comprehensive data quality initiative using SQL and Python, detecting 3,000 inconsistencies monthly and sterilizing impacted datasets.
- Implemented a star schema model across billing systems using Dimensional Modeling approaches, enhancing analytical outcomes for operational efficiency.
- Automated postal mail data processing pipelines through Python scripting, doubling throughput and reducing processing latency.
- Authored technical documentation for data migration, facilitating a seamless transition to cloud-based solutions without operational downtime.
- Deployed monitoring scripts through Azure Logic Apps, ensuring proactive anomaly detection and real-time alerting.

Junior Data Engineer

June 2016 – January 2018

QuillBot, Remote

- Built real-time data capture mechanisms using Azure Stream Analytics, processing 500,000 unique user requests daily with latency improvements from 5s to 500ms.
- Automated text analytics workflows with Python, consolidating linguistic data streams, reducing manual processing efforts by 2 hours daily.
- Constructed a robust metadata repository using SQL Server, promoting data accessibility and insight generation for over 200K text datasets.
- Initiated a cloud migration plan, successfully transitioning 50 on-premise servers to Azure, increasing infrastructure reliability significantly uptime.

EDUCATION

University of Texas at Austin, Austin, TX

April 2016

Master of Science in Computer Science

Advanced Data Structures • Cloud Computing • Machine Learning

SKILLS & OTHER

Data Engineering:	Microsoft Azure, Azure Data Factory, Azure Databricks, Spark, SQL, JSON
Data Architecture:	Data lakes, Data warehouse design, Dimensional modeling, Star schema
Automation & Integration:	OAuth 2.0, Data pipelines, Metadata management, Automation
Project Leadership:	Technical mentorship, Cross-functional collaboration, Project planning
Analytics & Monitoring:	Data quality management, Scalability, Reliability
Cloud Platforms:	Azure Data Lake, Synapse Analytics, Cloud data management